AGRISENSE

Precision Agriculture Crop Yield & Pest Risk Intelligence via Multimodal Sensor Fusion and GIS Decision Support

Institution: ABES Engineering College, Ghaziabad	**Academic Session:** 2026–2027
Department: Department of Information Technology	**Subject Code:** Major Project Review-2 (BCS 753)
Project Guide: Prof. Ayush Agarwal (Assistant Professor, IT)	**Target Publication:** IEEE Conference / Transactions Standard
Team Members: 1. Harish Kumar (Roll No: 2300320310086) 2. Devesh Singh (Roll No: 2300320130088) 3. Divyanshu Kumar Rai (Roll No: 2300320130100)	**Core Stack:** • ML: LightGBM (R^2=0.93), Random Forest (91%) • Full-Stack: React 18, TypeScript, Node.js, Leaflet GIS • AI/APIs: Gemini 1.5 Flash, HuggingFace, OpenWeather

1. Executive Project Summary

AgriSense is a dual-task precision agriculture intelligence platform designed to eliminate the historical disconnect between crop yield prediction and pest/disease outbreak prevention. Traditional precision farming solutions suffer from single-modality blindness (relying solely on regional macro-weather or static yield tables) and lack field-level actionability. AgriSense resolves this by ingesting **6 heterogeneous data streams** (weather telemetry, soil chemistry, Sentinel-2 satellite NDVI/EVI, Growing Degree Days phenology, crop metadata, and GPS field boundaries), aligning them into unified spatiotemporal feature vectors, and processing them through a decoupled dual-ML pipeline:

• **Crop Yield Regression:** LightGBM regressor achieving an empirical **R^2 = 0.93**, MAE = 0.27, and RMSE = 0.38.

• **Pest & Disease Risk Classification:** Random Forest ensemble classifier achieving **91% Accuracy**, 90% Precision, and 0.89 F1-score across 3 localized severity tiers.

• **Actionable GIS & Extension Hub:** Leaflet mapping with field pins, APMC Mandi revenue valuation engine, HuggingFace vision disease scanner, bilingual Malayalam/English/Hindi voice assistant (Web Speech API + Gemini 1.5 Flash), and an Agricultural Officer real-time escalation portal.

2. 3-Member Responsibility & Presentation Allocation Matrix

Member	Assigned Domain	Presentation Slides	Key Deliverables & Live Tasks
Member 1 **Harish Kumar** **(2300320310086)**	Team Lead & ML Data Ingestion Lead	**Slides 1, 2, 3, 4, 5** (Intro to Fusion Methodology)	• Project problem statement & 4 research gaps • 5 core technical project objectives • 6-phase multimodal data fusion pipeline • Agronomic GDD formula & feature alignment
Member 2 **Devesh Singh** **(2300320130088)**	Full-Stack & GIS Systems Engineer	**Slides 6, 8, 9, 13** (+ Live Software Demo)	• 4-tier system architecture & modular stack • Functional progress status audit • **Executes live UI demonstration** (GIS, Voice, Mandi, Officer Portal) • Technical challenges & caching/guardrail mitigations
Member 3 **Divyanshu Rai** **(2300320130100)**	ML Benchmarks & Research Lead	**Slides 7, 10, 11, 12, 14, 15** (+ Viva Q&A; Lead)	• Zero-leakage Blocked Spatial K-Fold protocol • Table II yield & pest classification benchmarks • IEEE format research paper submission status • Remaining roadmap (SHAP, ablation, Docker)

3. Complete Slide-by-Slide Speaking Script & Timing Guide

Follow this precise script during the 12-minute presentation. Each member speaks for approximately 3.5 to 4 minutes.

Part I: Member 1 (Harish Kumar) — Problem, Scope & Multimodal Pipeline

- **Slide 1 (Title):** Start with formal academic greeting. State project title: 'AgriSense: Precision Agriculture Crop Yield and Pest Risk Intelligence'. Introduce guide Prof. Ayush Agarwal and team members.
- **Slide 2 (Contents):** Briefly state the roadmap: Problem statement, 5 objectives, multimodal fusion pipeline, 4-tier architecture, empirical benchmarks, and live demonstration.
- **Slide 3 (Problem Statement):** Highlight 4 critical research gaps: 1) Single-modality blindness (weather alone fails to capture plant health), 2) Micro-climate blind spots (district averages hide field moisture stress), 3) Disconnected prediction tasks (treating yield and pest risk as isolated models causes conflicting advice), and 4) Lack of field actionability for smallholders.
- **Slide 4 (Project Objectives):** Present the 5 technical aims: 1) Fusing 6 heterogeneous data streams, 2) Building LightGBM yield regressor (R^2 >= 0.90), 3) Training Random Forest pest classifier (90%+ accuracy), 4) Engineering GDD thermal indices, and 5) Integrating GIS maps, voice AI, and officer escalation.
- **Slide 5 (Updated Methodology):** Explain the 6-phase multimodal ingestion and fusion architecture. Detail how weather, soil N-P-K, Sentinel-2 NDVI bands, and GDD are synchronized into uniform coordinate vectors using Savitzky-Golay cloud filtering. Hand over: 'I now invite Member 2 to present our system architecture and live working demonstration.'

Part II: Member 2 (Devesh Singh) — Architecture, Live Demo & Mitigations

- **Slide 6 (System Architecture):** Explain the 4-tier modular engineering: Layer 1 (Client: React 18, TypeScript, Tailwind, Leaflet GIS, Web Speech API), Layer 2 (Application: Node.js, Express REST API, Socket.io live chat, JWT), Layer 3 (AI & ML: LightGBM, Random Forest, Gemini 1.5 Flash, HuggingFace Vision), Layer 4 (Data: MongoDB Atlas with GeoJSON indexes, OpenWeatherMap, APMC Mandi).
- **Slide 8 (Implementation Progress):** Walk through the status audit: Core modules completed and functional (Leaflet GIS, Voice Assistant, Disease Scanner, APMC Mandi, Officer Portal); In-progress items (SHAP explainability console and ablation experiments).
- **Slide 9 & LIVE SOFTWARE DEMO:** Switch smoothly to browser at localhost:5173. 1) Open Leaflet GIS field map with GPS boundary pins; 2) Show live weather telemetry (28 deg C, 65% humidity) and GDD tracking; 3) Demonstrate leaf disease diagnosis with instant bio-chemical remedy; 4) Trigger bilingual voice assistant; 5) Switch to Officer Dashboard showing live farmer query escalation; 6) Show APMC Mandi valuation (e.g. 12.68 Tons -> Rs. 3.80 Lakhs).
- **Slide 13 (Challenges & Solutions):** Address real engineering bottlenecks: 1) Data heterogeneity solved by daily temporal resampling and spatial interpolation; 2) Leakage solved by Blocked Spatial K-Fold; 3) LLM hallucination solved by agronomic prompt guardrails; 4) API quotas solved by 6-hour caching TTL. Hand over: 'Now Member 3 will present our experimental validation, benchmarks, and IEEE paper.'

Part III: Member 3 (Divyanshu Rai) — Benchmarks, Paper & Future Scope

- **Slide 7 (Dataset & Setup):** Explain 6 input modalities, daily temporal granularity, 5-day Sentinel-2 passes, and the critical zero-leakage protocol: Blocked Spatial K-Fold (grouping farms by taluk) and out-of-season temporal holdout to guarantee forward generalization.
- **Slide 10 (Preliminary Results):** Deliver the winning metrics: LightGBM achieved R^2 = 0.93, MAE = 0.27, RMSE = 0.38 (outperforming XGBoost by 25% lower error due to leaf-wise histogram tree growth). Random Forest

achieved 91% accuracy, 90% precision, and 0.89 F1-score for pest classification due to ensemble bagging resilience against noisy field sensors.

- **Slide 11 (Research Paper):** State paper title: 'AgriSense: Precision Agriculture Crop Yield & Pest Risk Intelligence via Multimodal Sensor Fusion and GIS Decision Support'. Target: IEEE format. Mention the 4 claimed architectural contributions.
- **Slide 12 (Roadmap):** Outline remaining phases: 1) Multi-district dataset expansion, 2) Leave-one-modality-out ablation study, 3) SHAP feature attribution in UI, 4) Krishi Kendra field trials, 5) Production Docker containerization.
- **Slide 14 & 15 (References & Q&A;):** Cite 13 IEEE journal references (Klompenburg, Breiman, Ke et al.). Formally close: 'Thank you for your time. We are now open for questions.'

4. Empirical Results & Comparative Benchmark Tables

These exact empirical values from Slide 10 and your IEEE manuscript prove the superiority of your dual-model architecture.

Table A: Crop Yield Regression Benchmark (Continuous Metric Tons / Hectare)

Model Architecture	Split Strategy	R^2 Score	MAE	RMSE	Key Performance Rationale
LightGBM Regressor (AgriSense)	Blocked Spatial K-Fold	**0.93**	**0.27**	**0.38**	Leaf-wise histogram tree splitting on continuous multimodal features; lowest error.
XGBoost Regressor	Blocked Spatial K-Fold	0.89	0.36	0.49	Depth-wise tree growth shows higher error and longer training latency.
Random Forest Regressor	Blocked Spatial K-Fold	0.83	0.44	0.58	Bagging regressor struggles to extrapolate extreme climate anomalies.
Linear Regression Baseline	Blocked Spatial K-Fold	0.68	0.61	0.79	Fails completely due to non-linear interactions between weather, soil, and NDVI.

Table B: Pest & Disease Risk Multi-Class Classification Benchmark

Classifier Algorithm	Accuracy	Precision	Recall	F1-Score	Robustness & Architectural Behavior
Random Forest (AgriSense)	**91.0%**	**90.0%**	**89.0%**	**0.89**	Ensemble bagging (300 trees) excels on noisy humidity & leaf wetness sensors.
XGBoost Classifier	87.0%	86.0%	85.0%	0.85	Prone to over-emphasizing micro-sensor noise during boosting iterations.
Decision Tree (Single)	76.0%	74.0%	75.0%	0.74	High variance and frequent overfitting on isolated farm sensor anomalies.
Logistic Regression	71.0%	68.0%	70.0%	0.69	Underfits complex multi-class biological pathogen growth conditions.

5. Technical Architecture & Mathematical Formulations

1. Growing Degree Days (GDD) Formulation:

Calendar dates are poor indicators of biological crop growth due to seasonal temperature shifts. AgriSense computes cumulative thermal accumulation via GDD:

$$GDD = \max\left((T_{max} + T_{min}) / 2 - T_{base} , 0 \right)$$

where T_{base} is crop-specific (e.g., 10°C for maize/rice). This aligns satellite NDVI trajectories with biological maturity stages.

2. Satellite Signal Smoothing:

Sentinel-2 5-day optical passes suffer from cloud occlusions. AgriSense applies a **Savitzky-Golay digital filter** to reconstruct continuous, smooth 15-day NDVI trajectories:

NDVI = (Band 8 [NIR] - Band 4 [Red]) / (Band 8 [NIR] + Band 4 [Red])

3. Zero-Leakage Validation Protocol:

Standard random K-Fold cross-validation suffers from *spatial autocorrelation leakage*—farms within 5 km share climate and soil, yielding falsely high $R^2 > 0.98$. AgriSense enforces **Blocked Spatial K-Fold (k=5)** where whole taluks/clusters are isolated to either train or test, combined with **Out-of-Season Temporal Holdout**.

6. Top Evaluator Viva Voce Questions & Model Answers

Be prepared to answer these exact technical questions during the faculty defense:

Q1: Why choose LightGBM for yield instead of Deep Learning (LSTM/CNN)?

Answer: Crop yield at a field level is continuous tabular multimodal data (weather, NPK, NDVI). Deep learning models require millions of parameters and massive training sets to avoid severe overfitting, plus they suffer from high inference latency. LightGBM utilizes histogram-based gradient tree boosting with leaf-wise splitting, achieving an R^2 of 0.93 with minimal compute and deterministic feature explainability via TreeSHAP.

Q2: What is Data Leakage in agricultural ML, and how did you prevent it?

Answer: In agriculture, adjacent farms share identical atmospheric and soil conditions. If you perform random train-test splitting, data from neighboring fields leaks into the validation set, creating overly optimistic benchmarks (R^2 > 0.98) that fail on unobserved lands. We prevented this through Blocked Spatial K-Fold (grouping farms by taluk) and Temporal Out-of-Season Holdout.

Q3: Why are Yield Prediction and Pest Outbreak Risk solved together in AgriSense?

Answer: In classical literature, they are handled as disjoint pipelines. However, pest outbreaks cause biological canopy damage that directly degrades yield. Fusing them allows pest risk probability (e.g. 87% spore alert) to dynamically adjust expected yield forecasts and provide unified bio-chemical remedies rather than contradictory advice.

Q4: How does the Agricultural Extension Officer channel work technically?

Answer: When a farmer submits an escalation or pest probability exceeds 80%, our Express backend generates a GeoJSON query record in MongoDB Atlas. Socket.io emits a real-time event to the Zonal Officer Portal (OfficerQueriesView.tsx). The officer inspects the leaf image, reviews GIS coordinates, and returns verified prescriptions directly to the farmer.

Q5: How do you prevent Gemini AI from hallucinating incorrect pesticide advice?

Answer: We use domain-constrained prompt templates anchored to an expert agronomic dictionary and verified government chemical safety thresholds. The Gemini 1.5 Flash API is strictly restricted to safe dosages, bio-fungicide alternatives, and regional language translations via Web Speech API.

Q6: Why did Random Forest beat XGBoost in Pest Classification (91% vs 87%)?

Answer: Pest risk is a classification task operating on micro-climate sensor data (leaf wetness, humidity spikes). XGBoost boosts sequentially and can overfit to isolated sensor telemetry noise. Random Forest uses bagging over 300 decorrelated decision trees, which averages out noisy sensor variance and provides superior generalizability.

Q7: What is the status of your IEEE Research Paper?

Answer: Our paper titled 'AgriSense: Precision Agriculture Crop Yield & Pest Risk Intelligence via Multimodal Sensor Fusion and GIS Decision Support' is written in the standard IEEE 2-column format. It details our multimodal fusion pipeline, Table II benchmarks, and zero-leakage methodology. We are currently finalizing the ablation study before formal submission.

7. Live Software Demonstration Step-by-Step Runbook

- **Step 1: Open Farm GIS View:** Navigate to localhost:5173. Display the interactive Leaflet map canvas with field boundary polygons and nearest Krishi Seva Kendra pins.
- **Step 2: Inspect Live Field Telemetry:** Highlight the active weather dashboard: 28 deg C ambient temperature, 65% humidity, 8.2 km/h wind speed, and cumulative GDD tracking.
- **Step 3: Trigger Disease Leaf Vision Scanner:** Demonstrate plant leaf photo analysis powered by HuggingFace vision models. Point out the 87% outbreak alert and generated bio-chemical treatment recommendation.
- **Step 4: Demonstrate Bilingual Voice AI Assistant:** Click the microphone icon and speak in English or Malayalam. Show real-time Web Speech speech-to-text and contextual agronomic advice generated by Gemini 1.5 Flash.
- **Step 5: Switch to Officer Portal:** Log in to the Agricultural Officer Portal (OfficerDashboard.tsx / OfficerQueriesView.tsx) to showcase incoming farmer queries, spatial query distribution, and one-click resolution.
- **Step 6: Mandi APMC Revenue Simulator:** Display projected harvest yield (e.g. 12.68 Metric Tons) converted to current APMC market valuation (Rs. 3.80 Lakhs) for economic decision support.

8. Final Presentation Day Pre-Flight Checklist

✓ Backend server active on Port 3001 (Node.js & Express).
✓ Frontend Vite development server active on Port 5173 (React 18 & TypeScript).
✓ Two browser tabs pre-loaded: Farmer Dashboard and Agricultural Officer View.
✓ Presentation slides (AgriSense_Review2_FINAL_PRESENTATION.pptx) opened in fullscreen slide show mode.
✓ Microphone permissions enabled in browser for live Web Speech voice demo.
✓ Time limits strictly rehearsed: Member 1 (~3.5m), Member 2 (~4m + Demo), Member 3 (~3.5m). Total = 11 to 12 minutes.

Academic Defense Note: *This master dossier contains verified empirical data, architectural diagrams, and verbatim scripts corresponding to the Major Project Review-2 presentation at ABES Engineering College, Ghaziabad. Maintain strict coordination across all three team members.*